**MOTOR Ai Hiring Team**
*Scheringstraße 1*
*13355 Berlin, Germany*

October 16, 2026

Dear Hiring Team,

**Application for AI/ML Engineer (m/f/d) Autonomous Systems**

I am applying for the AI/ML Engineer position in Autonomous Systems at MOTOR Ai. I hold a master's degree in Vehicle Engineering from TU Berlin and currently work full-time at ZEKI. I want to bring my experience in model development, evaluation, and robotics integration to autonomous driving, contributing to your work toward deploying Level 4 systems.

In a visuo-tactile sensor-fusion project, I developed training and evaluation infrastructure for estimating in-hand object pose from RGB images and tactile measurements. I implemented and compared ten existing methods, then developed a transformer-based fusion model that matched the best evaluated baseline's pose-estimation accuracy. This work gave me experience translating published methods into working implementations and comparing models systematically. My technical toolkit includes Python, PyTorch, and TensorFlow.

At ZEKI, I developed a TurtleBot4 platform integrating SLAM, navigation, semantic mapping, and vision-language models. I also developed a Qwen-based agent that connects natural-language requests to navigation, exploration, and visual question answering. I am now developing a vision-language-action navigation policy integrated with Nav2, strengthening my experience connecting AI components to robot execution.

Alongside this work, I independently develop **TaoRobot: Autonomous Driving Full Stack**, an ongoing open-source personal project started in June 2026. I implement pose-graph SLAM, particle-filter localization, A* planning, and pure pursuit from scratch in lightweight ROS 2 nodes and integrate them in Gazebo. This gives me practical understanding of the autonomy stack into which learned models must fit. The code and demo are available at github.com/JinTTTT/taorobot.

My deployment experience also includes YOLO-based gesture detection on a real autonomous shuttle and GPU-accelerated visual SLAM on NVIDIA Jetson Xavier. These projects connect my AI work with vehicle behavior and constrained onboard computing.

I am available to start on December 1, 2026. Only the thesis remains in my second master's degree in Computational Engineering Science; I complete it outside working hours, and it will not affect my full-time availability. I would welcome the opportunity to discuss my contribution to your team.

Kind regards,

**Jin Tao**

**Jin Tao**
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